Customer Shopping Behavior Analysis

End-to-End Data Analytics Project Report

Tech Stack: Python (Pandas) · PostgreSQL · Power BI Dashboard

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1. Executive Summary

This project investigates transaction patterns and behavioral profiles across retail shoppers to uncover key drivers of revenue, customer retention, and category performance. Utilizing an end-to-end data pipeline comprising exploratory data analysis and feature engineering in **Python (Pandas)**, relational modeling and aggregations in **PostgreSQL**, and executive visualization in **Power BI**, raw customer transactions were synthesized into data-driven commercial strategies.

Total Cohort	Average Spend	Catalog Rating	Subscriber Rate
3,900 Customers	\$59.76 / order	3.75 / 5.00	27% (1,053 Users)

Non-subscribers constitute 73% of the buyer volume and yield the bulk of gross receipts, while Clothing and Accessories command over 75% of catalog demand.

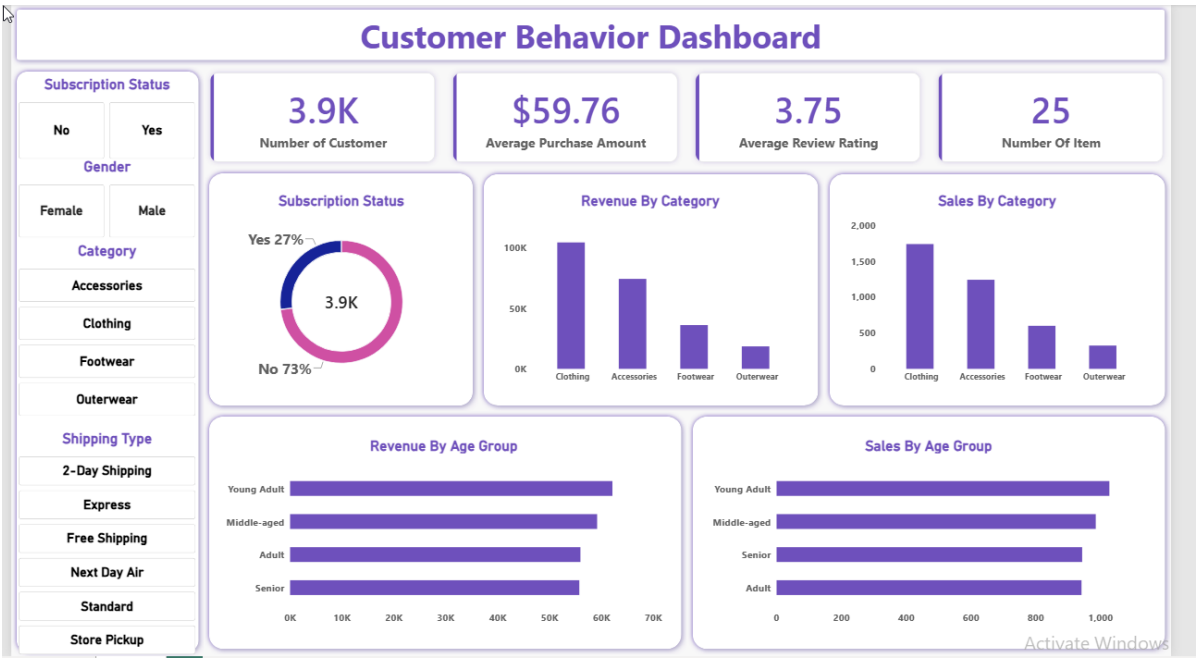
2. Technical Workflow & ETL Architecture

The project architecture followed a structured, repeatable analytical pipeline:

- **Data Cleaning & Imputation (Python):** Managed 37 missing records in `review_rating` by imputing category-specific median scores. Dropped the redundant `promo_code_used` attribute as it mirrored `discount_applied` identically. Normalized column identifiers into lowercase snake_case.
- **Feature Engineering:**
  - `age_group`: Partitioned continuous age distributions into four equal quartiles (*Young Adult*, *Adult*, *Middle-aged*, *Senior*) via `pd.qcut`.
  - `purchase_frequency_days`: Mapped qualitative cadence indicators (*Weekly* = 7, *Fortnightly* = 14, *Every 3 Months* = 90, *Annually* = 365) to quantitative intervals.
- **Relational Database Load:** Established an automated bridge via `SQLAlchemy` and `psycopg2` to persist the cleaned records directly into a local PostgreSQL schema table titled `customer`.
- **Analytical Reporting & Dashboarding:** Formulated SQL extraction queries and built interactive dashboard visuals for real-time exploratory decision support.

3. Power BI Dashboard Overview

Below is the visual dashboard representation highlighting category performance, age distributions, and subscription ratios:



Core Takeaways from the Visual Layer:

- **Category Dominance:** *Clothing* significantly surpasses all other lines in total revenue ($>$ \$100K) and item volume ($>$ 1,700 units), followed by *Accessories*, *Footwear*, and *Outerwear*.
- **Demographic Spread:** Revenue contributions across the four age cohorts remain remarkably balanced, with *Young Adults* exhibiting a minor lead in gross sales.

4. SQL Business Query Analysis & Solutions

Below are the queries executed against the PostgreSQL database along with operational findings:

Q1. Total Revenue Generated by Male vs. Female Customers

```
SELECT
    gender AS Gender,
    SUM(purchase_amount) AS Revenue_by_gender
FROM customer
GROUP BY gender
ORDER BY Revenue_by_gender DESC;
```

Analytical Finding: Male customers account for roughly 68% of total dataset transactions (2,652 entries), representing the primary revenue volume driver. Targeted marketing aimed at female demographic segments represents an untapped expansion opportunity.

Q2. Discount Users Spending Above the Catalog Average

```
SELECT
    customer_id, gender, item_purchased, category, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount > (SELECT AVG(purchase_amount) FROM customer)
ORDER BY purchase_amount DESC;
```

Analytical Finding: Pinpoints high-value, price-responsive buyers who maintain strong basket sizes despite promotional markdowns, protecting margin thresholds.

Q3. Top 5 Products by Highest Average Review Rating

```
SELECT
    item_purchased AS Top_five_rating_product,
    ROUND(AVG(review_rating)::NUMERIC, 3) AS Average_rating
FROM customer
GROUP BY item_purchased
ORDER BY Average_rating DESC
LIMIT 5;
```

Analytical Finding: Isolates flagship items boasting superior customer satisfaction for prominent display on landing pages and promotional banners.

Q4. Purchase Amount: Standard vs. Express Shipping

```
SELECT
    shipping_type AS Shipping_type,
    ROUND(AVG(purchase_amount)::NUMERIC, 2) AS Average_purchase
FROM customer
WHERE shipping_type IN ('Express', 'Standard')
GROUP BY shipping_type;
```

Analytical Finding: Ticket amounts are virtually indistinguishable between delivery tiers, showing that shipping upgrades correlate with fulfillment urgency rather than basket size.

Q5. Revenue and Spend Comparison: Subscribers vs. Non-Subscribers

```
SELECT
    subscription_status AS Subscription_status,
    COUNT(customer_id) AS Customer_count,
    ROUND(AVG(purchase_amount)::NUMERIC, 2) AS Average_purchase,
    ROUND(SUM(purchase_amount)::NUMERIC, 2) AS Total_purchase
FROM customer
GROUP BY subscription_status;
```

Analytical Finding: Average order sizes are comparable between groups; however, non-subscribers generate 73% of aggregate gross revenue due to absolute transaction volume.

Q6. Top 5 Products with Highest Discount Application Rate

```
SELECT
    item_purchased,
    COUNT(*) AS total_orders,
    ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0
        END) / COUNT(*), 2) AS discount_percentage
FROM customer
GROUP BY item_purchased
ORDER BY discount_percentage DESC
LIMIT 5;
```

Analytical Finding: Identifies stock keeping units (SKUs) vulnerable to promotional reliance, requiring pricing restructuring to preserve margins.

Q7. Customer Segmentation: New, Returning, and Loyal

```
SELECT
    customer_segment,
    COUNT(*) AS segment_count,
    ROUND(AVG(purchase_amount)::NUMERIC, 2) AS avg_segment_spend
FROM (
    SELECT
        customer_id,
        purchase_amount,
        CASE
            WHEN previous_purchases = 1 THEN 'New Customer'
            WHEN previous_purchases BETWEEN 2 AND 15 THEN 'Returning
                Customer'
            ELSE 'Loyal Customer'
        END AS customer_segment
    FROM customer
) sub
GROUP BY customer_segment
ORDER BY segment_count DESC;
```

Analytical Finding: Establishes concrete loyalty stages to orchestrate personalized email lifecycle cadences and retention workflows.

Q8. Top 3 Most Purchased Products Within Each Category

```
WITH RankedProducts AS (
    SELECT
        category,
        item_purchased,
        COUNT(*) AS total_sales,
        ROW_NUMBER() OVER(PARTITION BY category ORDER BY COUNT(*)
           DESC) AS item_rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT category, item_purchased, total_sales
FROM RankedProducts
WHERE item_rank <= 3;
```

Analytical Finding: Highlights category anchor items (e.g., Blouses and Sweaters in Clothing) to optimize inventory stocking cycles.

Q9. Repeat Buyers (>5 Previous Purchases) vs. Subscription Rate

```
SELECT
    subscription_status,
    COUNT(*) AS customer_count,
    ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER(), 2) AS
       percentage_share
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Analytical Finding: Unveils that frequent repeat buyers do not automatically subscribe, indicating that current loyalty incentives need improvement.

Q10. Revenue Contribution by Age Group

```
SELECT
    age_group,
    COUNT(customer_id) AS total_orders,
    ROUND(SUM(purchase_amount)::NUMERIC, 2) AS total_revenue,
    ROUND(100.0 * SUM(purchase_amount) / SUM(SUM(purchase_amount))
       OVER(), 2) AS revenue_pct
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Analytical Finding: Confirms balanced purchasing behavior across all age quartiles (∼24–26% each), indicating broad multi-generational appeal.

5. Strategic Recommendations

1. **Revamp Subscription Incentives:** With 73% unsubscribed and comparable basket amounts between cohorts, members need distinctive incentives (e.g., member-only bundles, threshold-free shipping).
2. **Cross-Sell Through Category Anchors:** Leverage high-volume Clothing lines (Blouses, Sweaters) as promotional traffic drivers to upsell accessories and footwear.
3. **Tiered Promotional Discounter Structure:** Transition from blanket promotions to minimum spend tiers (e.g., "$15 off on orders over $75") to elevate average order value above the $59.76 baseline.